

The Grand Cheat Sheet

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Dense, high-signal recall. For depth, open the per-topic file. Read top-to-bottom in ~20 min.

1 1. Concurrency & Multithreading

- › **Concurrency** $\neq$ **Parallelism**: concurrency = *dealing with* many things (structure); parallelism = *doing* many things at once (execution).
- › **Process** = own memory space; **Thread** = shared memory, own stack. Threads cheaper, but share state $\rightarrow$ bugs.
- › **Race condition**: outcome depends on timing of unsynchronized access to shared state. Fix with synchronization.
- › **Mutex** = lock, 1 owner (mutual exclusion). **Semaphore** = counter, N permits. **Binary semaphore** $\approx$ **mutex** (but no ownership). **Condition variable** = wait/signal on a predicate.
- › **Deadlock 4 conditions (must ALL hold)** $\rightarrow$ mnemonic "MHNC": **M**utual exclusion, **H**old & wait, **N**o preemption, **C**ircular wait. Break any one to prevent. (Order locks globally $\rightarrow$ kills circular wait.)
- › **Livelock** = threads act but no progress; **Starvation** = thread never scheduled.
- › **Producer-Consumer**: bounded buffer + 2 semaphores (empty, full) + mutex. **Reader-Writer**: many readers OR one writer.
- › **CAS / lock-free**: atomic compare-and-swap, no locks; **ABA problem** lurks. **volatile** = visibility (not atomicity).
- › **Thread pool**: reuse N worker threads from a queue $\rightarrow$ avoid thread-creation cost; bound concurrency.
- › **Takeaway**: "Shared mutable state + no synchronization = bug." Prefer immutability / message-passing.

2 2. Operating Systems

- › **Process vs Thread**: process = isolation + own address space; thread = shared address space, lighter context switch.
- › **Scheduling**: FCFS (convoy effect), SJF (optimal avg wait, needs prediction), **Round Robin** (fair, time-slice), Priority (starvation $\rightarrow$ aging), **MLFQ** (adaptive), Linux **CFS** (fair via vrun-time).
- › **Virtual memory**: every process sees its own contiguous address space; MMU maps virtual $\rightarrow$ physical via **page tables** (+**TLB** cache).
- › **Paging**: fixed-size pages; **page fault** $\rightarrow$ load from disk. **Replacement**: Optimal > LRU > Clock > FIFO (Belady's anomaly = FIFO can worsen with more frames).
- › **Thrashing**: too many page faults, CPU stalls on I/O $\rightarrow$ fix with working-set model / reduce multiprogramming.
- › **Context switch**: save/restore registers + PC + memory maps; expensive (flush TLB). Threads cheaper than processes.
- › **IPC**: pipes, shared memory (fastest), message queues, sockets. **fork()** = copy process (**copy-on-write**); **exec()** = replace image.
- › **File system**: **inode** = file metadata + block pointers; **journaling** = crash consistency.
- › **Takeaway**: kernel/user mode + system calls = the privilege boundary; virtual memory = the great illusion.

3. Computer Networks

- › **OSI** (mnemonic *"Please Do Not Throw Sausage Pizza Away"*): Physical, Data-link, Network, Transport, Session, Presentation, Application.
- › **TCP vs UDP**: TCP = reliable, ordered, connection (3-way handshake **SYN**→**SYN-ACK**→**ACK**), flow+congestion control. UDP = fast, fire-and-forget (video, DNS, games).
- › **TLS handshake**: ClientHello → ServerHello+cert → key exchange → finished. **Asymmetric to exchange a symmetric session key**, then symmetric for speed. TLS 1.3 = 1-RTT.
- › **DNS resolution**: browser cache → OS → resolver → root → TLD (.com) → authoritative → IP.
- › **REST** (stateless, resources, HTTP verbs, cacheable) vs **gRPC** (HTTP/2, protobuf, streaming, fast internal RPC) vs **GraphQL** (client picks fields).
- › **HTTP/1.1** (head-of-line blocking) → **HTTP/2** (multiplexing, server push, binary) → **HTTP/3** (QUIC over UDP, no TCP HOL blocking).
- › **WebSocket** = full-duplex persistent; **SSE** = server→client stream; **long-polling** = fallback.
- › **L4 LB** (transport, IP/port, fast) vs **L7 LB** (application, content-aware routing).
- › **Idempotency**: GET/PUT/DELETE idempotent; POST not.
- › **"Type google.com → ?"**: DNS → TCP handshake → TLS → HTTP GET → server → render. Know every hop.

4. Cloud & Infrastructure

- › **Container vs VM**: container shares host kernel (namespaces=isolation, cgroups=limits), MBs, secs; VM = full OS, GBs, mins.
- › **Docker image** = layered (union FS), immutable; container = running instance.
- › **Kubernetes**: control plane (**API server, etcd, scheduler, controller-mgr**) + nodes (**kubelet, kube-proxy**). Pod = smallest unit; Deployment = replicas+rollout; Service = stable virtual IP; Ingress = L7 routing.
- › **Autoscaling**: HPA (pods by metric), VPA (resize), Cluster Autoscaler (nodes).
- › **Service mesh** (Istio/Envoy sidecar): mTLS, retries, traffic-splitting, observability — off the app code.
- › **CI/CD deploys**: **Rolling** (gradual), **Blue-Green** (instant switch, easy rollback), **Canary** (small % first).
- › **IaC**: Terraform (declarative, multi-cloud) vs CloudFormation (AWS).
- › **Observability = 3 pillars**: **Metrics** (Prometheus, aggregate), **Logs** (events), **Traces** (request across services). **SLI** (measured) → **SLO** (target) → **SLA** (contract+penalty); **error budget** = 1–SLO.
- › K8s = "desired state" reconciliation loop. You declare; controllers converge.

5. System Design

- › **Framework**: Requirements (functional + non-functional) → Estimation → API → Data model → High-level diagram → Deep dive → Bottlenecks/trade-offs.
- › **Scale**: **Vertical** (bigger box, limit + SPOF) vs **Horizontal** (more boxes, needs statelessness + LB). **Stateless** scales easily.
- › **Caching strategies**: **Cache-aside** (lazy, app manages), **Write-through** (write cache+db together), **Write-back** (write cache, async db — fast but risky), **Write-around**. Eviction: **LRU/LFU**. Hardest problem = **invalidation**.

- › **CDN**: cache static content at edge, near users.
- › **Sharding** (partition data horizontally; watch hot partitions) + **Replication** (copies; leader-follower = read scaling, multi-leader = write availability).
- › **CAP**: on partition, pick **C** (reject) or **A** (serve stale). CA only without partitions (single node). **PACELC**: else, latency vs consistency.
- › **Message queues** (Kafka/SQS/RabbitMQ): decouple, buffer, smooth spikes, async. **Event-driven** = react to events, loose coupling.
- › **Microservices**: independent deploy/scale; cost = distributed complexity, network, data consistency.
- › **Latency numbers**: L1≈1ns, RAM≈100ns, SSD≈100μs, disk seek≈10ms, network RT (same DC)≈0.5ms, cross-continent≈150ms.
- › Always state assumptions + estimates + trade-offs out loud. Silence loses points.

6. Low-Level Design (LLD)

- › **OOP 4 pillars** (mnemonic "**A PIE**"): **A**bstraction, **P**olymorphism, **I**nheritance, **E**ncapsulation.
- › **SOLID**:
 - **Single Responsibility** — one reason to change.
 - **Open/Closed** — open to extend, closed to modify.
 - **Liskov Substitution** — subtype usable as base type.
 - **Interface Segregation** — small focused interfaces.
 - **Dependency Inversion** — depend on abstractions, not concretions.
- › **Composition > Inheritance** ("has-a" > "is-a"); low coupling, high cohesion; DRY/KISS/YAGNI.
- › **Patterns** (by category):
 - **Creational**: Singleton, Factory, Abstract Factory, Builder, Prototype.
 - **Structural**: Adapter, Decorator, Facade, Proxy, Composite.
 - **Behavioral**: Strategy, Observer, Command, State, Template Method, Iterator.
- › **UML relationships**: inheritance (←--), composition (◆--), aggregation (◇--), association (--), dependency (- ->).
- › LLD approach: clarify → nouns=classes, verbs=methods → relationships → apply patterns → code core → discuss extensibility. Don't over-pattern.

7. Distributed Systems

- › **8 Fallacies**: network is reliable, latency=0, bandwidth ∞, network secure, topology stable, one admin, transport cost=0, network homogeneous — *all false*.
- › **CAP** → see #5. **Consistency models**: Strong > Linearizable > Causal > Read-your-writes > Eventual.
- › **Consistent hashing**: hash nodes+keys onto a ring; adding/removing a node moves only ~K/N keys. **Virtual nodes** smooth load.
- › **Quorum**: $R + W > N$ → strong consistency (overlap guarantees latest read). $W=N$ (durable), $R=1$ (fast reads).
- › **Consensus** (agree on one value despite failures): **Raft** = leader election (terms, votes) + log replication + safety (commit when majority). **Paxos** = correct but notoriously hard to grok. Used by etcd/ZooKeeper/Spanner.
- › **Leader election**: avoid **split-brain** with quorum + fencing tokens.
- › **Vector clocks / Lamport timestamps**: order events / detect concurrent updates.

- › **2PC** (blocking, coordinator SPOF) vs **Saga** (local txns + compensations, eventual).
- › **Delivery**: at-most-once, at-least-once (retries→dedup needed), "**exactly-once**" = **at-least-once** + **idempotency** (true E2E is a myth).
- › **Kafka**: topic → partitions (ordered, parallel) → consumer groups; offset-based, replayable, durable log.
- › Real systems: DynamoDB/Cassandra = AP/tunable; Spanner = CP (TrueTime); etcd/ZK = consensus.

8. Databases

- › **ACID**: Atomicity (all-or-nothing), Consistency (valid state), Isolation (concurrent = serial-ish), Durability (survives crash).
- › **Isolation levels** × **anomalies**:

Level	Dirty Read	Non-Repeatable	Phantom
Read Uncommitted	possible		
Read Committed			
Repeatable Read			(MySQL: via MVCC)
Serializable			

- › **Indexes**: **B+tree** (range queries, reads, RDBMS) vs **LSM-tree** (write-heavy, Cassandra/RocksDB; memtable→SSTables→compaction). **Clustered** = data sorted by key (1/table); **non-clustered** = separate pointer structure.
- › **Joins**: INNER (both), LEFT (all left + matches), RIGHT, FULL, CROSS (cartesian).
- › **CTE** = WITH × AS (...) readable/recursive. **Window functions** = OVER(PARTITION BY ... ORDER BY ...) (ROW_NUMBER, RANK, LAG/LEAD) — aggregate *without* collapsing rows.
- › **Locking**: **Optimistic** (version check at commit, low contention) vs **Pessimistic** (lock up-front, high contention). **MVCC** = readers don't block writers.
- › **SQL vs NoSQL**: SQL = relations, ACID, joins, vertical. NoSQL families: **KV** (Redis), **Document** (Mongo), **Column-family** (Cassandra), **Graph** (Neo4j) — denormalize, scale horizontally, BASE.
- › Index = faster reads, slower writes + storage. Don't index everything.

9. Performance Engineering

- › **Latency** (time/request) vs **Throughput** (requests/sec) vs **Bandwidth** (max capacity). Optimize for the one that matters.
- › **Tail latency**: track **p50/p90/p99/p999** — averages lie. At scale, p99 hits most users on a fan-out request (Google "Tail at Scale").
- › **Little's Law**: $L = \lambda \times W$ (items-in-system = arrival-rate × time-in-system). **Amdahl's Law**: speedup capped by serial fraction.
- › **USE** (resources: Utilization, Saturation, Errors) + **RED** (services: Rate, Errors, Duration).
- › **Bound types**: CPU-bound (high util, profile hot path), I/O-bound (waiting, batch/async), memory-bound (cache misses, GC), network-bound.
- › **Memory hierarchy** (cache locality matters): register→L1(1ns)→L2→L3→RAM(100ns)→SSD(100µs)→c
- › **Workflow**: **Measure** → **Profile (flame graph)** → **find bottleneck** → **fix ONE thing** → **re-measure**. "Premature optimization is the root of all evil."

- › **Memory leak:** heap grows unbounded; find via heap dumps / allocation profiling.
- › **Speed levers:** caching, batching, connection pooling, pipelining, async, reducing N+1.
- › "Service got 10× slower" → check recent deploys, metrics (USE/RED), profile, narrow CPU vs I/O vs lock contention vs downstream.

10. Security Fundamentals

- › **AuthN** = *who are you* (identity) vs **AuthZ** = *what can you do* (permissions). AuthN before AuthZ.
- › **CIA triad:** Confidentiality, Integrity, Availability.
- › **Encryption vs Hashing vs Encoding:** encryption = reversible w/ key (confidentiality); hashing = one-way (integrity, passwords); encoding = not security (Base64).
- › **Symmetric** (AES, one shared key, fast, bulk data) vs **Asymmetric** (RSA/ECC, public/private, key exchange + signatures, slow).
- › **Password storage:** hash + **unique salt** with **bcrypt/argon2/scrypt** (slow on purpose). Never plaintext/MD5.
- › **OAuth 2.0** = delegated *authorization* (give app access without password); use **Authorization Code** + **PKCE** for apps. **OIDC** adds *authentication* (ID token) on top. **JWT** = header.payload.signature — signed (not encrypted by default!); verify signature, check exp, don't store secrets in payload.
- › **OWASP Top 10** (know a few cold): Broken Access Control (#1), Cryptographic Failures, **Injection** (SQLi → parameterized queries), Insecure Design, Misconfiguration, Vulnerable Components, Auth Failures, Data Integrity, Logging Failures, SSRF.
- › **Web attacks:** **XSS** (inject script → escape output/CSP), **CSRF** (forged request → anti-CSRF token/SameSite), **SQLi** (→ prepared statements).
- › **Principles:** least privilege, defense in depth, zero trust, secrets in **Vault/KMS** (never in code/env-in-git), rotate secrets.
- › "Design secure login": TLS + hashed/salted passwords + rate limiting + MFA + secure session/JWT + least privilege.

11. Behavioral / Leadership / Googleyness

- › **STAR:** Situation (context, brief) → Task (your responsibility) → Action (what **YOU** did, 60%) → Result (**quantified** impact).
- › Use **"I" not "we"** for your actions. Always end with metrics ("reduced latency 40%", "saved \$200k/yr").
- › **Themes & what they probe:** Leadership (drive without authority), Ownership (end-to-end, beyond your scope), Conflict (disagree respectfully, data-driven, **disagree & commit**), Collaboration (cross-team influence), Ambiguity (act with incomplete info), Mentoring (grow others), Impact (measurable outcomes).
- › **Amazon LPs** (bar raiser): Customer Obsession, Ownership, Invent & Simplify, Dive Deep, Have Backbone; Disagree & Commit, Deliver Results, etc. Map a story to each.
- › **Google "Googleyness":** comfort with ambiguity, collaboration, intellectual humility, doing the right thing.
- › **Story bank:** prep **6–8 stories** that flex across themes (1 conflict, 1 failure, 1 leadership/impact, 1 ambiguity, 1 mentoring, 1 cross-team). Build a story × theme matrix.
- › **Failure question:** pick a *real* failure, own it, show what you *learned/changed*. Never blame others; never a fake-humble "I work too hard."
- › Top mistakes: rambling, no metrics, "we" instead of "I", no failure story, blaming others. Be concise + structured + quantified.

12 Cross-Topic Memory Hooks

Connection	Hook
OS ↔ Concurrency	Threads/processes/scheduling/locks are OS primitives concurrency builds on
Networks ↔ System Design	LB, caching, CDN, protocols = the plumbing of every design
Databases ↔ Distributed Systems	CAP, replication, sharding, quorum, consistency overlap heavily
Cloud ↔ Distributed Systems	K8s/service-mesh = distributed-systems patterns productized
Performance ↔ everything	Profiling + tail latency + caching cut across OS, DB, network
Security ↔ Networks/System Design	TLS, auth, secrets are non-functional requirements in every design

13 The 12 Highest-Frequency Interview Concepts (memorize cold)

1. CAP theorem + consistency models
2. TCP handshake + TLS + "type a URL" flow
3. Caching strategies + invalidation
4. Sharding + replication + consistent hashing
5. ACID + isolation levels matrix
6. B+tree vs LSM-tree indexes
7. Deadlock 4 conditions ("MHNC")
8. Process vs thread + virtual memory/paging
9. SOLID + composition over inheritance
10. Tail latency (p99) + measure-before-optimize
11. OAuth/JWT + OWASP injection/XSS/CSRF
12. STAR with quantified "I" results

Golden rule for every round: state assumptions → think out loud → give the trade-off, not just the answer.